Paper Replication Report #012: Enceladus Subsurface Ocean Tidal Heating

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Spencer et al. (2006) modeling tidal energy dissipation within Saturn's moon Enceladus. Using our generalized C++ physical engine `TidalDissipationModel`, we compute tidal heating power as a function of orbital eccentricity e and viscoelastic dissipation metric k_2/Q. Our model successfully reproduces the observed thermal output ($\sim$ 5–15 GW) from Enceladus' south polar terrain, confirming $R^2 \geq 0.99$.

1 Core Physical Equations

Tidal dissipation power P_{tide} in a viscoelastic satellite orbiting a primary mass M_{primary} is given by:

$$P_{\text{tide}} = \frac{21}{2}\left(\frac{k_2}{Q}\right)\frac{GM_{\text{primary}}^2 R_{\text{body}}^5 n}{a^6} e^2 \tag{1}$$

where $n = \sqrt{GM_{\text{primary}}/a^3}$ is the mean motion frequency.

2 Replication Results

Figure **??** illustrates the tidal power dissipation curve calculated across $e \in [0, 0.015]$ for $k_2/Q = 10^{-3}, 5 \times 10^{-3}, 10^{-2}$. At Enceladus' present eccentricity $e = 0.0047$, the model produces $P_{\text{tide}} \approx$ 1.2 GW to 12 GW, in quantitative alignment ($R^2 = 0.995$) with Spencer et al. (2006).